Higher National Diploma in Electrical Engineering

4TH Year, First Semester Examination – 2018

EE 3204- Electrical Energy Management.

Instructions for Candidates:
Answer any five questions.
All questions carry equal marks.
Calculators are allowed.

No of pages: 03
No of questions: 06
Time: Three (03) hours

01.

I) Explain the construction and working of a sodium vapor lamp. (05 Marks)

ii) List down five factors to consider when designing a lighting system? (05 Marks)

iii) An office has length of 20m, width of 10m, height of 3m. Ceiling to desk height is 2m, Area be illuminated to a general is 250 lux using twin lamp 32 Watt ,CFL Luminaire with a SHR of 1.25. Each lamp has an initial output of 85 lumen per watt. Maintenance factor is 0.63, Utilisation factor is 0.69. Design a lighting system for this office

(10 Marks)

Total (20 Marks)

02.

I) Write 5 objectives of a proper tariff system (05 Marks)

ii) Explain in detail the block rate tariff system and two part tariff system. (05 Marks)

iii) Explain how synchronous motors can be use to improve the power factor of a system.

(10 Marks)

Total (20 Marks)

03.

I) Explain the working principle of three phase induction motor using relevant vector diagrams. (05 Marks)

ii) Explain in detail how the speed of a three phase induction motor can vary using change of stator voltage and frequency. (05 Marks)

iii) The rotor of a six pole, 50 HZ slip ring IM has a resistance of 0.2 ohms per phase and runs at 960 rpm on full load. Calculate the resistance per phase to insert in the rotor circuit such that the speed is reduced to 800 rpm, the torque being the same as before. (10 Marks)

Total (20 Marks)

04.

I) What is Magnetising current of a transformer? (05 Marks)

ii) Draw a neat phasor diagram and explain the working of a practical transformer on no load condition. (05 Marks)

iii) The no load current of a transformer is 10A at a power factor of 0.25 lagging, when connected to 400V, 50 Hz supply. Calculate,

a) Magnetising component of no load current.

b) Iron loss and Maximum value of flux in core (10 Marks)

Total (20 Marks)

05.

I) Explain Voltage Regulation of a transformer. (05 Marks)

ii) Derive an equation for voltage regulation of a transformer stating all the quantities in the equation. (05 Marks)

iii) A 250/125 V, 5 kVA single phase transformer has primary resistance of 0.2 ohm and reactance of 0.75 ohm. The secondary resistance is 0.05 ohm and reactance of 0.2 ohm.

Determine its voltage regulation while supplying full load on 0.8 power factor leading.

The secondary terminal voltage on full load and 0.8 power factor leading. (10 Marks)

Total (20 Marks

06.

I) Explain the importance of power factor improvements. (04 Marks)

ii) List down the purpose of Supply side management and Demand side management (06 Marks)

iii) A single-phase motor takes a current of 12A at 0.75 power factor lagging from 220V, 50 Hz supply. Find the capacitance to improve the power factor to 0.9 lagging.

(10 Marks)

Total (20 Marks)